

DiaTool-DPO: Enhancing Tool-Augmented LLM Dialogue via Direct Preference Optimization

Multi-turn preference optimization with automatic paired trajectory construction

For ML researchers and NLP engineers working on tool-augmented LLMs and RLHF alignment

State 1 Initial

State 2 Tool+Missing Slots

State 3 Tool+Complete Slots

State 4 Wait Tool Response

State 5 Complete

SFT Alone Fails to Teach Slot-Filling and Tool Rejection

Why supervised imitation is insufficient for multi-turn tool-augmented dialogue control

Observed in TA-LLMs trained with SFT-only: incomplete argument handling and out-of-scope rejection are under-optimized.

Failure 1

No follow-up questions for missing slots

Model skips clarification and proceeds without collecting required arguments.

Failure 2

Hallucinated tool invocation

Model issues tool calls with fabricated/missing slot values.

Failure 3

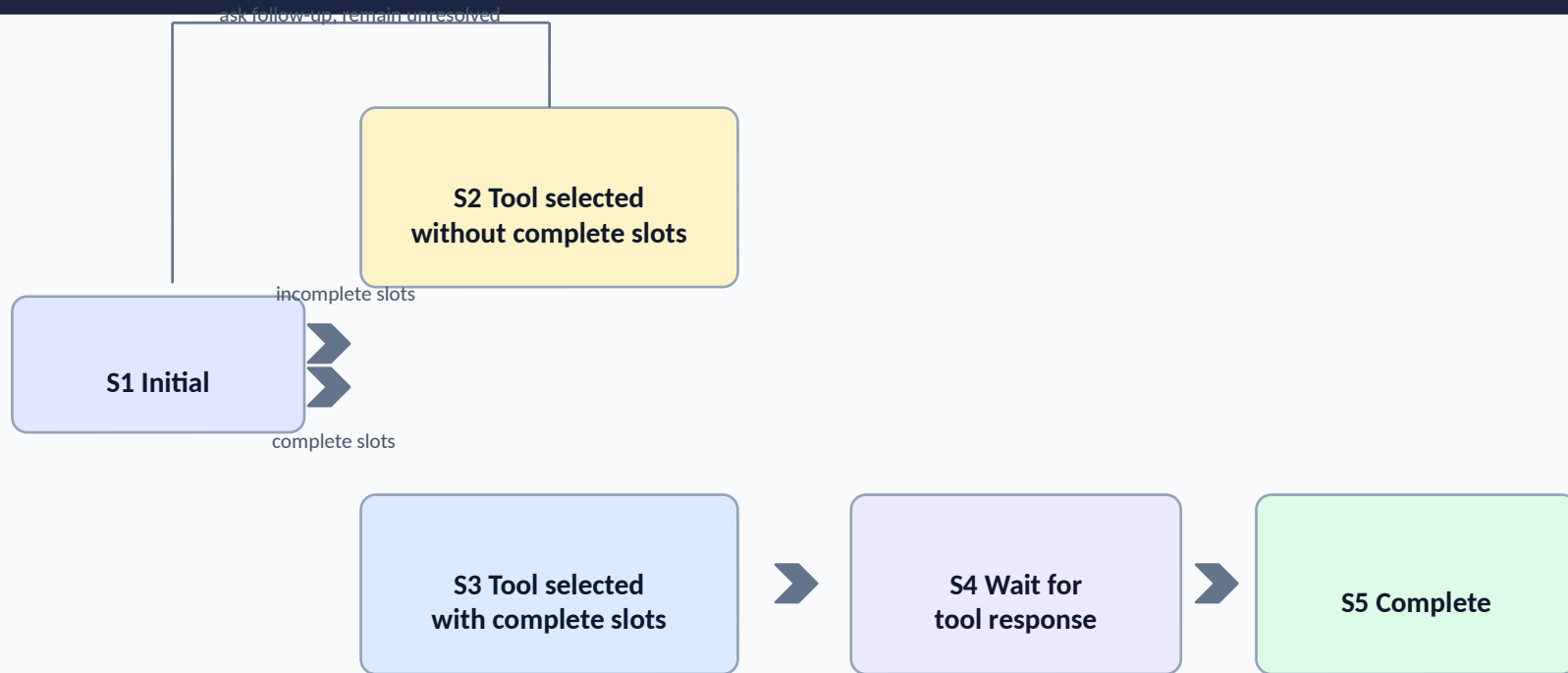
No rejection for improper requests

Model fails to reject out-of-scope or inappropriate tool-call requests.

Motivation: Introduce preference optimization over dialogue trajectories to explicitly reward proper slot-filling and tool-call rejection behavior.

Modeling TA-LLM Dialogue as a 5-State Markov Decision Process

State abstraction foundation for systematic preference construction



5-state MDP from Figure 1 enables trajectory-level policy comparison instead of single-turn imitation, forming the basis of DiaTool-DPO.

Three Query Types Define Distinct State Trajectories

Compact mapping from query semantics to expected state transitions

Type 1		

Automatic Paired Trajectory Construction Without Human Labels

Chosen vs. rejected trajectories derived from MDP state expectations

Type 1: chosen = direct tool execution after complete slots;
rejected = unnecessary follow-up / delayed call

Type 2: chosen = ask follow-up until slots complete; rejected =
slot hallucination / premature tool call

Type 3: chosen = reject tool call and stay in S1; rejected =
forced tool selection despite mismatch

Type 1 trajectories	315	Low
Type 2 trajectories	408	High
Type 3 trajectories	275	Medium
Total	998	-

Learning lessons:

- Prevent redundant slot-filling
- Prevent slot hallucination
- Learn proper tool call reject/accept

SFT + DiaTool-DPO Achieves Best Macro Average Across All Models

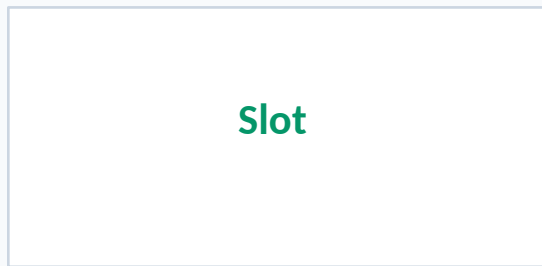
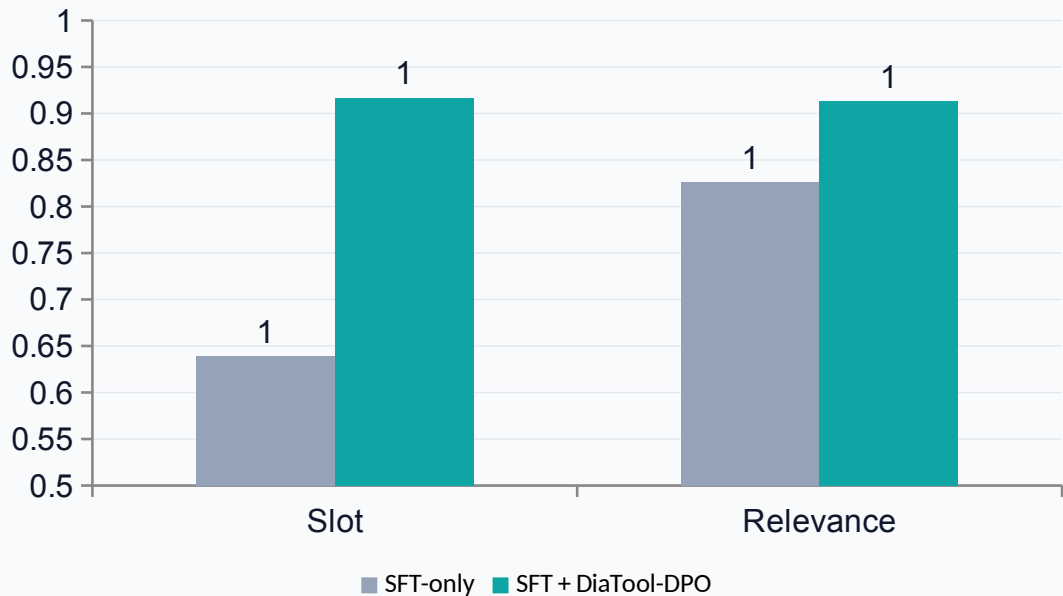
Main benchmark comparison: Call, Completion, Slot, Relevance, Micro Avg, Macro Avg

Prop.-8B	DiaTool-DPO-only	0.529	0.817	0.444	0.783	0.646	0.643
Prop.-8B	SFT-only	0.900	0.916	0.694	0.913	0.870	0.856
Prop.-8B	SFT + DiaTool-DPO	0.886	0.929	0.833	0.826	0.884	0.868
Prop.-3.1B	DiaTool-DPO-only	0.357	0.551	0.528	0.391	0.455	0.457
Prop.-3.1B	SFT-only	0.771	0.817	0.750	0.826	0.790	0.791
Prop.-3.1B	SFT + DiaTool-DPO	0.743	0.871	0.833	0.826	0.765	0.818
LLaMA-3-8B	DiaTool-DPO-only	0.029	0.449	0.056	0.261	0.205	0.199
LLaMA-3-8B	SFT-only	0.843	0.957	0.639	0.826	0.844	0.816
LLaMA-3-8B	SFT + DiaTool-DPO	0.857	0.929	0.917	0.913	0.905	0.904

Across all model sizes, SFT + DiaTool-DPO delivers the best balanced Macro Avg (0.868 / 0.818 / 0.904).

LLaMA-3-8B: +43.5% Slot Accuracy and +10.5% Relevance with DiaTool-DPO

Before/after gains when adding DiaTool-DPO on top of SFT



Key Takeaways: RL-Based Dialogue Control for Production TA-LLMs

- 5-state MDP formulation enables systematic trajectory pairing for preference learning.
- Automatic chosen/rejected pair construction removes human-label dependency.
- SFT + DiaTool-DPO consistently improves balanced performance (Macro Avg).
- DPO alone is insufficient; it complements, not replaces, supervised training.
- Approach is language-agnostic and reaches near GPT-4o-level conversational tool use.

Deployment implication: use RL-based dialogue control to improve slot completeness and safe tool-call rejection in production TA-LLMs.